

Energy inflation in the shadow of US–China tensions: Is the energy transition a pathway to stability?

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ABSTRACT

Energy is the backbone of today's economy and is considered the engine of growth in the contemporary world. Almost 80% of the energy requirements are still fulfilled by fossil fuels, the prices of which are vulnerable to external shocks. Geopolitical risks stemming from rising US–China tensions can serve as a catalyst for energy inflation, whereas the transition to clean energy sources can help mitigate the implications of energy inflation. Therefore, the primary objective of this analysis is to examine the influence of US–China tensions and the energy transition on energy inflation, utilizing the novel method of moments quantile regression (MMQR) estimator for the period of 1995–2023. The results indicate that increasing US–China tensions lead to elevated energy inflation across all quantiles. Conversely, energy transition proves significant in mitigating energy inflation at lower and medium levels. The advancement of technology and globalization has an adverse effect on energy inflation across all quantiles, indicating that both elements contribute to a decrease in energy costs at low, medium, and high levels. The findings suggest that governments should prioritize energy transition policies as a strategic tool to mitigate energy inflation in the context of rising US–China geopolitical tensions.

1. Introduction

Energy inflation refers to persistent increases in the prices of energy goods and services, including oil, gas, and electricity, typically measured using the energy component of the consumer price index (OECD, 2026). The significance of energy inflation has increased in recent years, particularly after the global economy experienced a sharp return to inflation following a long period of relatively stable prices. The IMF (2022) reports that global inflation rose from 4.7% in 2021 to 8.8% in 2022, before easing to 6.9% in 2023 and to 4.1% by 2024. Globally, the inflation surge has been strongly energy-driven. The World Bank shows that its energy price index rose by 50% between January 2020 and December 2021, and then increased by a further 26.3% between January and April 2022, reflecting sharp jumps in oil, coal, and natural gas prices. The World Bank also reports that in nominal terms, crude oil prices increased by about 350% from April 2020 to April 2022, while coal and gas prices reached historic highs (World Bank, 2022). During the same period, the US–China tensions raised geopolitical risk and

policy uncertainty, creating inflationary pressures due to a higher risk premium and disruptions to trade and supply chains (Zaghdoudi, 2026), which ultimately have sizeable effects on oil-market outcomes. On the other hand, energy transition efforts can slow down the pace of energy inflation over time by reducing fossil-fuel dependence and lowering wholesale electricity prices as renewables expand (Cevik and Ninomiya, 2023).

The role of energy has, indeed, become central in driving economic growth in the contemporary world. It is necessary for the success of all economic endeavours because without energy, it is almost impossible to manufacture products and services. Consequently, the economic development of a nation largely depends on its capacity to produce affordable and sustainable energy. Urbanization and industrialization are important indicators of economic expansion, and both processes eventually lead to increased energy consumption. Therefore, economic growth and energy requirements within an economy are typically linked to one another (Sadorsky, 2009). Industrialization was accelerated by the post-World War II economic recovery, which facilitated the extraction

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and utilization of fossil fuels on a large scale (Gross et al., 2022). Fossil fuels, the primary energy source for contemporary industry and living, have become the primary source of worldwide energy consumption. Fossil fuels accounted for over 84% of global primary energy in 2019 (Ritchie and Roser, 2020). Although fossil fuels remain the dominant source of energy in most nations, their prices are highly vulnerable to global events, such as tensions between the US and China, international efforts to transition to renewable energy, and instability in energy markets, resulting in severe implications for economic outcomes. In the past few years, trade conflicts, especially between the US and China, energy price changes, political unrest, and oil market volatility have all contributed to the destabilization of the world's economic system (Saint Akadiri and Ozkan, 2025). Both energy-importing and energy-exporting countries are affected by these events, which also disrupt market equilibrium and exacerbate general financial instability. Strategic behavior by major fossil-fuel producers can strengthen the link between US–China tensions and energy inflation. When geopolitical tensions rise, uncertainty also increases, and oil markets become more sensitive to supply news. Producer groups such as OPEC and OPEC + can respond strategically by cutting or slowing supply growth to sustain revenues, thereby tightening global supply and pushing prices up (Kilian, 2009). At the same time, US–China tensions play a crucial role in shaping oil-market dynamics and prices (Cai et al., 2022).

The rise in trade and political tensions between the US and China has been one of the most pressing issues in recent times, heightening the uncertainty surrounding global trade (Steinbock, 2018). Due to trade barriers and taxes that alter global consumption trends, this prolonged debate has had a significant impact on energy markets. In addition, growing fiscal constraints and political conflicts have increased uncertainties around energy prices, leaving energy-importing nations at higher risk. Global shocks and policy frictions shape energy inflation. US–China geopolitical tensions raise uncertainty and risk premia in oil and gas markets, which spill over into higher prices, making energy markets a key channel to inflation (Pinchett, 2024). Trade wars and higher trade costs can raise inflation by increasing import prices and disrupting supply chains, including energy inputs (Cheng et al., 2023). By contrast, energy transition progress—especially higher renewable penetration—tends to lower wholesale electricity prices, easing energy inflationary pressures (Cevik and Ninomiya, 2023). Because these forces occur together, their combined effects can be nonlinear and stronger during high-inflation episodes.

The international oil market has seen significant changes during the last 20 years. Before China joined the WTO, the Chinese economy had a minimal impact on the oil market; however, the scenario changed altogether after China formally entered the WTO. Indeed, with 16.4% of the total worldwide oil use in 2021, China has emerged as a major stakeholder in the industry (BP, 2022). In terms of oil use and production, the US ranks first in the world, accounting for 19.9% of oil use and 16.8% of the world's oil production. The dynamics of oil prices might be significantly impacted by the development of these two nations' political ties, given their relative importance. Ever since Donald Trump became the US president, the tensions between China and the US have escalated manifold due to the trade war between the two economies, Hong Kong's judicial independence, intellectual property concerns, and Taiwan's international standing. According to Cai et al. (2022), the trade war between the US and China has led to a decline in Chinese output, as the US has imposed higher tariffs on Chinese imports, while China has increased taxes on a number of US imports. As a result of this decline in Chinese growth, China's Oil demand has significantly reduced. Due to a stable supply and a decline in demand, energy prices have become very volatile. This phenomenon, along with many others, underscores the significant impact of rising tensions between the US and China on energy inflation, as well as other sectors.

The volatile prices of conventional energy sources are not the only concern; their negative consequences for the ecosystem are also ringing alarm bells. A shift to more eco-friendly and green energy sources is

necessary for solving this problem. International organizations, including the United Nations and the OECD, have emphasized that nations should transition from traditional energy sources to low-carbon sources of energy. In addition to environmental concerns, the transition to renewable energy sources is influenced by energy inflation. The connection between energy inflation and energy transition is complicated and bidirectional, causing each other (Li et al., 2023; Nchofoung, 2024). On the one hand, rising inflation encourages people and companies to reduce their energy usage, purchase energy-efficient goods, and transition from fossil fuels to clean energy. Furthermore, rising energy prices encourage more investment in alternative energy sources. On the other hand, massive spending on renewable infrastructure and increasing demand for critical minerals may lead to higher energy costs, thereby contributing to inflationary pressure (D'orazio, 2024). However, as renewable energy sources expand and decrease in cost, reliance on conventional fossil fuel markets diminishes, potentially stabilizing or reducing energy expenses. Thus, the transition to clean energy sources is becoming increasingly appealing as a way to reduce reliance on oil and counter the shocks of rising energy inflation (Nchofoung, 2024). The available empirical literature has confirmed that higher energy prices can accelerate the transition to renewable energy sources and increase their consumption (Murshed and Tanha, 2021). However, the empirical evidence on the impact of energy transition on energy inflation is scarce.

The importance of the issue lies in the fact that the world economy remains very dependent on fossil fuels, the price of which is very sensitive to geopolitical conflicts, trade restrictions, and disruption of the supply chain. The global energy markets have been considered by uncertainties due to the growing tension between the two leading economies and energy consumers in the world, the USA and China, which has led to higher energy prices and inflation. Meanwhile, governments are more and more turning to renewable energy sources to enhance energy security and cut reliance on fossil fuels. The question of whether energy transition can help to reduce the volatility of energy prices in the context of geopolitical turmoil is therefore of significant importance for policymakers and energy actors. Since energy inflation is crucial in impacting the industrial sector, transportation, household consumption, and economic stability, examining the connection between these factors has relevance for the achievement of sustainable and stable energy systems on a global scale.

Given the long-lasting consequences of energy inflation for sustainable economic development, the literature on energy inflation is growing. However, most of the past studies have estimated its consequences for several macroeconomic and ecological factors, including environment (Li et al., 2020), economic growth (Dagoumas et al., 2020), and energy transition (Li et al., 2023), among others. Some of the past studies have investigated how energy prices can impact the prices of clean energy (Reboredo and Ugolini, 2018) and the ecosystem (Ulussever et al., 2025). Reboredo and Ugolini (2018) found that oil and electricity prices significantly influenced clean energy stock returns in the USA and Europe, but their analysis focused on financial market responses rather than inflationary pressures within the energy sector. Similarly, Ulussever et al. (2025) demonstrated that energy price indices and crude oil prices worsen environmental degradation in GCC countries, while geopolitical risks exert mixed effects depending on country conditions. Although these studies recognized the importance of energy prices and geopolitical factors, neither investigates how these forces translate into energy inflation dynamics. Another strand of literature examined the relationship between oil prices and the energy transition process (Murshed and Tanha, 2021; Nchofoung, 2024). Murshed and Tanha (2021) reported that rising oil prices can stimulate renewable energy consumption, but only after oil prices exceed a threshold level. Likewise, Nchofoung (2024) showed that oil price shocks hinder energy transition, particularly in oil-exporting economies, while Jin and Kim (2023) argued that renewable energy consumption can hedge against oil price risks in OECD economies. All these studies treated energy transition as a dependent variable and overlooked its potential role in

influencing energy inflation.

Studies on the determinants of energy inflation have mostly been linked to geopolitical risks (Olasehinde-Williams et al., 2024). Olasehinde-Williams et al. (2024) confirmed that geopolitical risks increase energy inflation in European economies. In the past few years, the prevalence of geopolitical threats worldwide has altered the energy supply-demand dynamics, generating significant interest among authorities and investors. Due to the considerable influence of geopolitical risks on energy markets, various research efforts have investigated the relationship between these risks and energy markets (Mignon and Saadaoui, 2024; Su et al., 2025), as well as their effect on energy transitions (He et al., 2025). Likewise, Mignon & Saadaoui (2024) showed that geopolitical risks and improved US–China political relations both affect oil prices through expectations and supply concerns, while Su et al. (2025) highlighted that geopolitical risks significantly alter the volatility of clean and traditional energy markets. Furthermore, He et al. (2025) demonstrated that geopolitical risks can even accelerate renewable energy transition under certain structural conditions. More recently, Li et al. (2025) and Zaghdoudi et al. (2026) specifically examined US–China tensions and concluded that these tensions intensify oil price uncertainty and fossil fuel price volatility. Nevertheless, these studies mainly focused on energy market volatility and price uncertainty rather than energy inflation itself. Therefore, despite the growing literature on energy prices, geopolitical risks, and energy transition, two important gaps remain unresolved. First, the existing studies limited examine these relationships separately, without integrating US–China tensions and energy transition into a unified framework for explaining energy inflation. Second, prior research has mainly focused on oil price volatility, renewable energy adoption, or financial market reactions, while the direct influence of US–China tensions and energy transition on energy inflation has received little empirical attention. To address these gaps, the present study investigates the relationship among US–China tensions, energy transition, and energy inflation.

This analysis aims to address the aforementioned gaps and makes the following contributions to the existing body of knowledge. First, the research assesses the impact of the US–China tensions on energy inflation. This relationship highlights how rising global tensions may increase energy prices by restricting energy trade and hindering energy supplies. Second, examining the relationship between energy transition and energy inflation within the geopolitical context is another significant contribution of the analysis, highlighting the role of increased consumption and production of renewable energy sources in mitigating the impact of energy inflation in an era marked by geopolitical tensions. Third, the study contributes by integrating US–China geopolitical risk and energy transition within a unified quantile-based framework, offering a clearer understanding of how these factors impact on energy inflation. Fourth, estimating the impact of US–China tensions and energy transition on the energy inflation dynamics of 33 global economies is another novel contribution of the analysis. These 33 economies include both developed and developing economies, which are important in shaping the global energy dynamics. Moreover, estimating the nexus in these economies offers more valuable information regarding the response of energy inflation to rising US–China tensions and energy transition in varying social, economic, and political frameworks. Fifth, the study employs the novel MMQR method that helps us estimate the asymmetric impact of US–China tensions and energy transition on energy inflation across its various levels. Asymmetric analysis offers more valuable information, and this is justified as most macro variables are asymmetric in nature due to the involvement of human behavior, as is the case with our variables. Moreover, these variables follow the asymmetric pattern and non-normal behavior; therefore, we must use the nonlinear modelling while estimating the nexus between energy inflation, US–China tensions, and energy transition. Sixth, the MMQR also helps uncover heterogeneity across quantiles, showing that the impact of US–China tensions strengthens at higher levels of energy inflation, while the stabilizing role of energy transition is more effective

at lower to medium quantiles. This shifts the focus from average effects to distributional dynamics, providing more policy-relevant insights. Lastly, this research is timely as it provides practical recommendations for governments, international institutions, and energy-sector stakeholders to mitigate energy inflation.

2. Theoretical framework

The rising tensions between the US and China have significantly increased the global geopolitical risk that can directly contribute to energy inflation through several mechanisms. First, increasing trade barriers or imposing sanctions can hinder global energy chains, significantly increasing logistics and compliance expenses, which results in higher prices for energy goods and services (Omokaro et al., 2025). Secondly, the perception of higher uncertainty and risk gives rise to a geopolitical premium in commodity markets. Investors want protection against higher risk and seek compensation for potential supply disruptions, thereby incorporating additional costs into energy prices. Third, to counter rising risks and improve energy security, countries may focus on storing oil and gas, which could further restrict the global energy supply and potentially lead to higher prices (Umbach, 2010). Fourth, increased competition for acquiring critical energy and mineral technologies can slow the progress of the transition towards affordable, low-carbon energy sources, thereby increasing reliance on unstable and vulnerable fossil fuel markets. Fifth, due to higher geopolitical risks, governments and businesses anticipate any price increase before it occurs and start purchasing and storing energy, thereby increasing demand and enhancing energy prices prior to the actual onset of physical shortage. Lastly, due to increased competition in currency and settlement systems, transaction risks and exchange rate expenses associated with global energy trade have risen, resulting in higher energy prices (Mirza et al., 2023). Ultimately, we can conclude that rising US–China tensions increase geopolitical risks, which can exacerbate energy inflation by hindering the energy supply chain and increasing the cost of energy trade. Thus, we propose the following hypothesis 1: *Rising US–China tensions increase energy inflation.*

The energy transition can impact energy inflation in both the short-term and long-term. In the short term, the transition to clean and green energy sources can lead to increased energy prices. Renewable energy is widely acknowledged as a low-carbon energy source that can improve the standard of living (Lin and Li, 2022); however, the higher cost is attached to higher living standards due to the high capital and technological expenses required for deployment. According to the US consumer price index, the prices of energy-related goods and services increased more between 2021 and 2022 than any other items, mainly attributed to the higher cost of transitioning to renewable energy sources. However, the higher costs associated with integrating renewables are often temporary, due to infrastructure development, technology adoption, and supply chain adjustments. This happens because the integration of clean and low-carbon energy within the energy system initially raises costs but can later reduce inflation through energy savings, technological efficiency, and reduced environmental damage in the long run. Since renewable energy is less vulnerable to external shocks and geopolitical risks, substituting conventional energy sources with renewable energy helps stabilize prices (Su et al., 2021), thereby reducing long-term energy inflation. Thus, while the energy transition may initially exacerbate inflationary pressures, it ultimately brings stability to energy prices and provides a sustainable supply in the long run, thereby helping to mitigate energy inflation. On the basis of the above discussion, we proposed the following hypothesis 2: *Energy transition reduces energy inflation in the long run.*

Rising US–China tensions increase global geopolitical risk, which raises energy inflation mainly through cost and risk channels. Trade restrictions, sanctions, and supply chain disruptions increase transportation, compliance, and financing costs in global energy markets, which are passed on to energy prices (Yang and Fu, 2025). At the same

time, higher uncertainty creates a geopolitical risk premium, as investors and traders demand compensation for possible supply disruptions. Precautionary behaviors, such as storing oil and gas by governments and firms, further squeeze supply and intensify price pressures. In contrast, the energy transition works as a stabilizing force over time. Although shifting to renewable energy may raise costs in the short run due to high investment and infrastructure needs, these effects are largely temporary. As renewable energy capacity expands, energy systems become less dependent on fossil fuels and less exposed to geopolitical shocks, helping to stabilize energy prices in the long run (Fouquet, 2016). Together, this framework explains why geopolitical tensions tend to increase energy inflation, while the energy transition gradually offsets these pressures by improving energy security and price stability, though its effects may weaken during periods of extreme inflation. The theoretical framework flowchart is presented in Fig. 1.

3. Econometric model and methodology

The theoretical basis of this study is linked to the geopolitical tensions as a cause of energy market uncertainty, energy supply chain disruption, rising energy costs, and a contributor to energy inflation. Meanwhile, energy transition can strengthen the energy sector and mitigate geopolitical shocks. To empirically validate these hypotheses, the study used a panel data set of 33 countries from 1995 to 2023, selected based on data availability. This study assesses the role of US–China tensions and energy transition in energy inflation, alongside economic growth, financial development, technology development, urban population, and globalization. Following the works of Attilio (2025) and Li et al. (2025), the model used in this study is presented as:

$$EI_{it} = \kappa_0 + \kappa_1 UCT_{it} + \kappa_2 ET_{it} + \kappa_3 EG_{it} + \kappa_4 FD_{it} + \kappa_5 TD_{it} + \kappa_6 UP_{it} + \kappa_7 GLOB_{it} + \epsilon_{it} \quad (1)$$

In the above model, energy inflation (EI) is the function of US-China tension (UCT), energy transition (ET), economic growth (EG), financial development (FD), technology development (TD), urban population (UP), and globalization (GLOB). For the other factors: κ_0 stands for intercept, ϵ represents the error term, and $i(t)$ denotes the countries

(years). Panel data is most often influenced by cross-sectional dependence (CD), which is a primary concern while dealing with panel analysis. This issue mainly arises due to the correlation between all units within a cross-section. As per the study of Pesaran (2004), the correlation between the cross-sectional units is attributed to the following elements, such as unobserved common variables, geographical effects, and spillover effects evolving from socioeconomic connections between these units. Since its existence can give rise to contradictory results and erroneous conclusions, tackling CD is essential while dealing the panel data (Pesaran, 2004). Therefore, checking the data for the existence of CD is critical. The baseline equation of the CD test by Pesaran is given below as:

$$CD_{test} = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{k=i+1}^N \hat{\tau}_{ik} \right) \quad (2)$$

In homogeneous panel data frameworks, it is assumed that all cross-sectional units or nations share the same slope parameter. Heterogeneous models permit individual variations in some or all model parameters. The panel data vary across countries and time periods, suggesting that this type of data is quite diverse. In this situation, assuming a homogeneous slope may yield incorrect or unreliable estimates. Consequently, this work used the Pesaran and Yamagata (2008) test to assess the slope homogeneity phenomenon. To avoid spurious results, this study also employs Im-Pesaran-Shin (IPS) unit root test (Im et al., 2003). This test is used to examine the stationarity of the variables and to determine their order of integration, which is essential for empirical analysis. In order to see whether a genuine long-term relationship exists between the variables (EI, UCT, ET, EG, FD, TD, UP, GLOB) containing a unit root that alters across different panel units, countries in our case. The Kao (1999) and Pedroni (1999) methods are among the most efficient cointegration methods for estimating a reliable, sustainable relationship among the above-stated variables (EI, UCT, ET, EG, FD, TD, UP, GLOB). The null hypothesis of these tests assumes no cointegration, and the alternative hypothesis suggests otherwise.

This study uses the Driscoll-Kraay fixed effects (DK-FE), two-stage

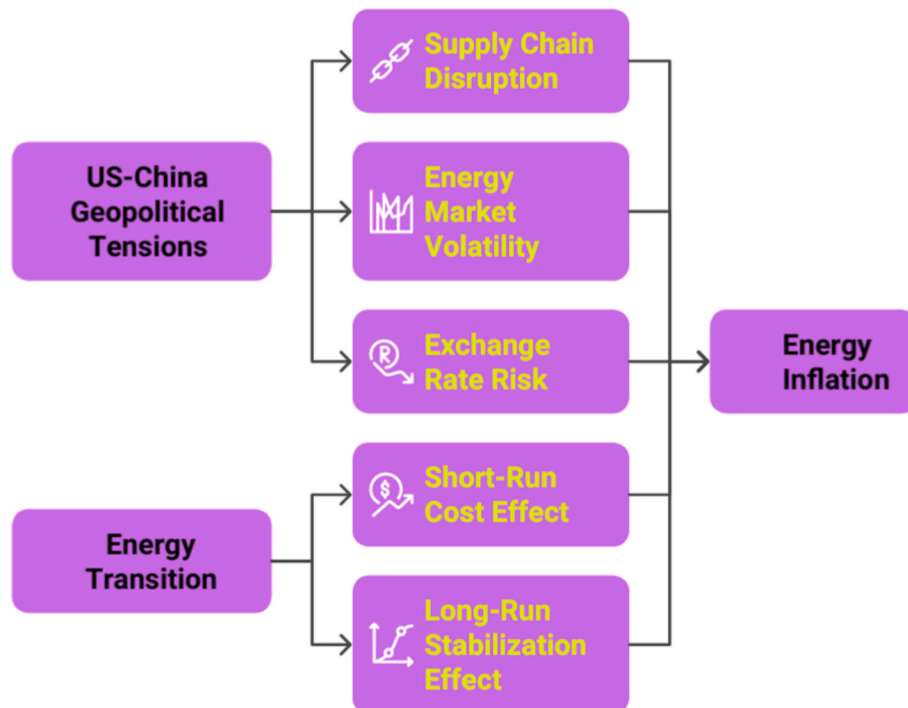


Fig. 1. Theoretical framework flowchart.

least squares (2SLS), Lewbel 2SLS, and system generalized method of moments (SGMM) methods as baseline estimators for the panel data analysis. The use of the DK-FE estimator is especially suitable when there is heteroskedasticity in the data, as it mitigates issues associated with possibly variable variance across panels (Driscoll and Kraay, 1998). Utilizing heteroskedasticity-robust standard errors enhances the reliability of our analysis findings, providing acceptable inferences despite the violation of the homoskedasticity criterion. This method improves the reliability of our results, enabling more precise statistical conclusions amid differing degrees of heteroskedasticity among panels. This approach has several advantages, as it overcomes the issues of unobserved heterogeneity across countries, cross-sectional dependence, heteroskedasticity, and serial correlation while producing robust standard errors. However, DK-FE cannot control endogeneity, which is a serious concern in panel analysis. Thus, to control for endogeneity, we have employed the 2SLS and the Lewbel 2SLS method. These methods ensure consistent estimation by instrumenting the endogenous variable, with the Lewbel approach providing additional internally generated instruments when strong external instruments are unavailable.

The 2SLS method is used when an explanatory variable is correlated with the error term, which can cause biased results in OLS. This issue normally arises due to omitted variables, measurement errors, or simultaneity. 2SLS neutralizes this problem by using instrumental variables (Cumby et al., 1983). In the first stage, the problematic variable is predicted using valid instruments that are related to it but not directly linked to the outcome variable. In the second stage, this predicted value is used to estimate the final relationship, helping to correct endogeneity and produce more reliable coefficients. The Lewbel (2012) 2SLS approach is a special case of instrumental-variable estimation designed for situations where traditional external instruments are weak or unavailable. Instead of relying on outside instruments, this method creates internal instruments by exploiting heteroskedasticity in the data. By using higher moments of the model errors, it constructs instruments that are uncorrelated with the disturbance term but correlated with the endogenous regressor. This makes Lewbel 2SLS especially useful in macroeconomic and energy studies where finding strong external instruments is difficult.

To capture the dynamic relationship between the variables, we employed SGMM, which further supports our estimates by mitigating endogeneity concerns using lagged variables as instruments. This is particularly suitable for panel data with potential persistence. We use the SGMM estimators designed by Arellano and Bond (1991), originally presented by Holtz-Eakin et al. (1988), which is an instrumental variable technique. Ullah et al. (2018) state that the GMM model is reliable, efficient, and conforms to normalcy due to its utilization of information incorporated into the moment circumstances. Furthermore, the framework assists in alleviating the endogeneity issue arising from the connection between the lagged outcome variable and the error component. Endogeneity arises when factors mutually impact one another, hence confounding their causal links and resulting in biased outcomes. Endogeneity arises from omitted variables, simultaneity, autocorrelation errors, and measurement errors. This study's implementation of the GMM model aligns with current research, including Lau et al. (2023), and Wang et al. (2022). Specifically, we use the two-step approach GMM, which accommodates endogenous covariates. The SGMM has shown robustness with small sample sizes, exhibiting optimal characteristics for small-sample bias (Roodman, 2009). The reliability of the system estimator depends on many assumptions, including the independence of error terms, the legitimacy of instruments, and the lack of connection between variations in supplementary instruments and fixed country-specific effects. These methods are commonly used in the energy economics literature (Da Silva and Cerqueira, 2017; Berk et al., 2020). The literature confirms that current energy inflation is significantly influenced by the previous year's energy inflation. Therefore, this variable is included, and the system GMM model is specified as follows:

$$\begin{aligned} EI_{it} = & \kappa_0 + \lambda_1 EI_{it} + \kappa_1 UCT_{it} + \kappa_2 ET_{it} + \kappa_3 EG_{it} + \kappa_4 FD_{it} + \kappa_5 TD_{it} + \kappa_6 UP_{it} \\ & + \kappa_7 GLOB_{it} + v_i + \epsilon_{it} \end{aligned} \quad (3)$$

We have performed some tests to verify the robustness of our estimates. The Hansen (1982) tests for over-identifying limitations were employed to examine the overall efficiency of the instruments and determine the reliability of the estimates from the two-step SGMM framework. The null of this test states that instruments are exogenous, meaning they are valid. The Arellano–Bond autocorrelation test is conducted on the initial differences to assess autocorrelation for AR(2). The null hypothesis posits the absence of autocorrelation.

3.1. MMQR

To capture distributional heterogeneity, the current study used a unique estimating approach, namely the MMQR test, as the main estimation approach, which was essentially created by the research of Machado and Silva (2019). When compared to other regression procedures, MMQR uses contemporaneous criteria, such as enforcing distribution assumptions or confirming the existence of the moment function, to obtain results. Because it takes into account the conditional heterogeneous covariance resulting from the elements of the endogenous regressors, the MMQR approach is the most popular. The MMQR method uses multiple quantile analysis to show how the parameters are related. Evaluation of distributional and heterogeneous impacts across quantiles is possible using panel quantile regression (Hao and Naiman, 2007). Additionally, it accurately reflects real data on the correlation between the variables being studied, taking into account the steady influence of distributional variations. As a result, the MMQR shows distinct relationships between chosen parameters in various conditional distributions that are not achievable with conventional regressions that rely on mean variable estimates. In order to ascertain the distributive impact of the regressors on the outcome variable over different quantile ranges, it is crucial to estimate the tested variables at the conditional distribution. To estimate the conditional quantiles under the location–scale variant model, we begin with the following equations:

$$J_{it} = \varphi_{it} + K'_{it}\phi + (\chi_i + L_{it}\lambda')\epsilon_{it} \quad (4)$$

Country-specific fixed effects are denoted by $i = 1, 2, \dots, n$, while the dimension of time is represented by $t = 1, 2, \dots, t$. The term L refers to a K -dimensional vector containing the unidentified regressors. Using this information, we can redefine the above equation as:

$$Q_J(\tau / K_{it}) = (\varphi_i + \lambda_i q(\tau)) + K'_{it}\phi + L_{it}\lambda q(\tau) \quad (5)$$

In the above specification (5), the K_{it} is the vector representing the outcome variable. The function $Q_J(\tau / K_{it})$ is used to highlight the distribution of quantiles for the outcome variable that depends on the value of regressor K'_{it} . The equation $(\varphi_i + \lambda_i q(\tau))$ is the scalar coefficient that indicates the time-invariant influence of quantile τ for cross-section i . The smoothed instrumental variables quantile regression (SIVQR) method extends this analysis by correcting for endogeneity within the quantile framework. The SIVQR estimator (Kaplan, 2022) is applied when an important regressor is endogenous and the goal is to estimate effects at various points of the outcome distribution, not just the average. Compared with traditional quantile regression, the SIVQR estimator offers clearer and more reliable results when endogeneity is present. It uses instruments to correct bias caused by reverse causality or missing factors, which standard quantile regression cannot handle. At the same time, SIVQR still captures differences across low, medium, and high quantiles, allowing a better understanding of how effects vary across the outcome distribution.

To address endogeneity concerns and identify them, this analysis uses the energy transition as the potential endogenous variable, as it is

co-determined with energy inflation or influenced by omitted factors. In this analysis, our outcome variable is energy inflation, while other regressors, US–China tensions, economic growth, financial development, technological development, urban population, and globalization, are considered exogenous and are used as instruments under a general instrumentation strategy. To identify the instruments, we have employed the internal-instrumentation approach, in which each exogenous variable serves as an instrument. The instruments are assumed to be correlated with energy transition but not directly linked to the error term. Lewbel's 2SLS approach also employs internal instruments, particularly when external instruments are difficult to identify. Moreover, the SIVQR framework is used to account for endogeneity across different quantiles of energy inflation. This approach integrates the instrumental variable approach with quantile regression, producing reliable estimates while accounting for heterogeneous effects across the distribution. In the existing literature (Inglesi-Lotz et al., 2025), advanced quantile regression techniques are employed to examine the impact of the US–China tension index within a panel data framework. Thus, we have achieved identification through (i) the use of exogenous regressors as instruments, (ii) heteroskedasticity-based instruments via the Lewbel approach, and (iii) robust estimation using SIVQR, ensuring that the results are not biased due to endogeneity.

3.2. Data and descriptive analysis

To explore the effect of US–China tensions and energy transition on energy inflation, this study assembles data from 1995 to 2023. The sample of the study includes 33 global economies (see Table A1). The sample is limited to these countries because data on energy inflation were only available for these 33 global economies. Energy inflation represented by EI, is measured through the energy-based consumer price index and the data series is drawn from the OECD. US–China tension (UCT) is captured through US–China tension index, which is developed by Rogers et al. (2024). Energy transition (ET) is measured as the ratio of renewable energy consumption to fossil fuel consumption (% of total final energy consumption). The data series for renewable and fossil fuel energy consumption are extracted from the WDI. The authors compute the ET data series to capture the relative shift from fossil-based to renewable energy sources. The authors calculate this data series. Economic growth (EG), financial development (FD), technology development (TD), urban population (UP), and globalization (GLOB) are control variables in our model. Economic growth (EG) is represented by annual percent of GDP growth. Higher economic growth leads to increased energy demand as transportation, industrial production, and consumption rise, thus raising energy prices. However, in advanced economies, economic growth accompanied by technological innovation can moderate this effect. Hence, the linkage between economic growth and energy inflation can be both positive and negative, depending on the energy usage and the structure of the economy. Financial development (FD) is measured through the financial development index. Financial development facilitates investment in renewable energy technologies and infrastructure that can ease supply constraints and stabilize energy prices in the long term. On the other hand, financial development can also stimulate energy-intensive industrial expansion, thus enhancing energy prices in the short term. Technological development (TD) is proxied by the total number of patent applications from both residents and nonresidents. Technological development enhances energy efficiency and encourages the utilization of renewable energy sources, thereby mitigating energy inflation by reducing dependence on volatile fossil fuel markets (Solarin et al., 2022). Urban population (UP), measured by urban population as percent of the total population, represents the degree of urbanization. Urbanization tends to rise energy consumption due to increased demand for transportation, industrial output, and electricity, thereby increasing energy prices (Yu et al., 2020). Globalization (GLOB), proxied by the KOF Globalization Index, encompasses economic, political, and social dimensions of global

integration. Higher globalization can positively or negatively influence energy inflation. Economic growth, technology development, and urban population data is drawn from the WDI, while financial development data is sourced from the IMF. All variables are transformed into natural logarithms before estimation to ensure consistency. Table 1 displays the list of variable descriptions along with their definitions and data sources.

The descriptive statistics are displayed in Table 2. Mean values provide the average scores of each variable, while the maximum and minimum scores provide the range of variations within the dataset. The mean values for EI, UCT, ET, EG, FD, TD, UP, and GLOB are 1.662, 4.578, 0.415, 2.452, 0.619, 8.232, 4.334, and 4.362, respectively. The standard deviation scores range between 0.118 and 3.017. Skewness measures the symmetry of data distribution. It is reported that EI, EG, FD, UP, and GLOB exhibit negative skewness, confirming that these variables have left tails. In contrast, UCT, ET, and TD exhibit positive skewness, indicating that these variables have a right tail. Finally, the Jarque-Bera test examines the normality of the dataset. All variables are statistically significant, confirming the absence of normality of the dataset. Similar results are also found in the Q–Q plots (Fig. 2).

The correlation matrix and VIF results for EI, UCT, ET, EG, FD, TD, UP, and GLOB are displayed in Table 3. In a correlation matrix, the correlation coefficient ranges from -1 to 1 . A positive coefficient indicates a positive relationship, while a negative coefficient indicates a negative relationship. It is shown that ET, FD, TD, UP, and GLOB are negatively correlated with EI, while UCT and EG are positively correlated with EI. Additionally, no issue of high correlation is detected in our data set. The VIF scores range from 1.09 to 2.75, with the mean VIF 1.71, confirming the absence of multicollinearity in the data set.

4. Empirical results

As a preliminary step, we investigate the existence of CD among the variables. The outcomes of the CD test are reported in Table 4. The findings indicate that all variables are significant, leading to the rejection of the null hypothesis of no CD among all variables. This confirms the presence of CD among all variables under analysis. In the next step, slope heterogeneity is assessed using the Pesaran and Yamagata tests. All the variables are statistically significant except EG. It indicates that all variables have heterogeneous slopes, except for EG, which has a homogeneous slope in our model. This study also employs the IPS test, which effectively accounts for both CD and heterogeneity in panel data. The IPS test results are reported in Table 5. The findings indicate that for EI, UCT, ET, TD, and GLOB, the null hypothesis cannot be rejected at levels, suggesting that these variables become stationary after first differencing, i.e., they are integrated of order $I(1)$. In contrast, EG, FD, and UP are found to be stationary at levels, implying an order of integration

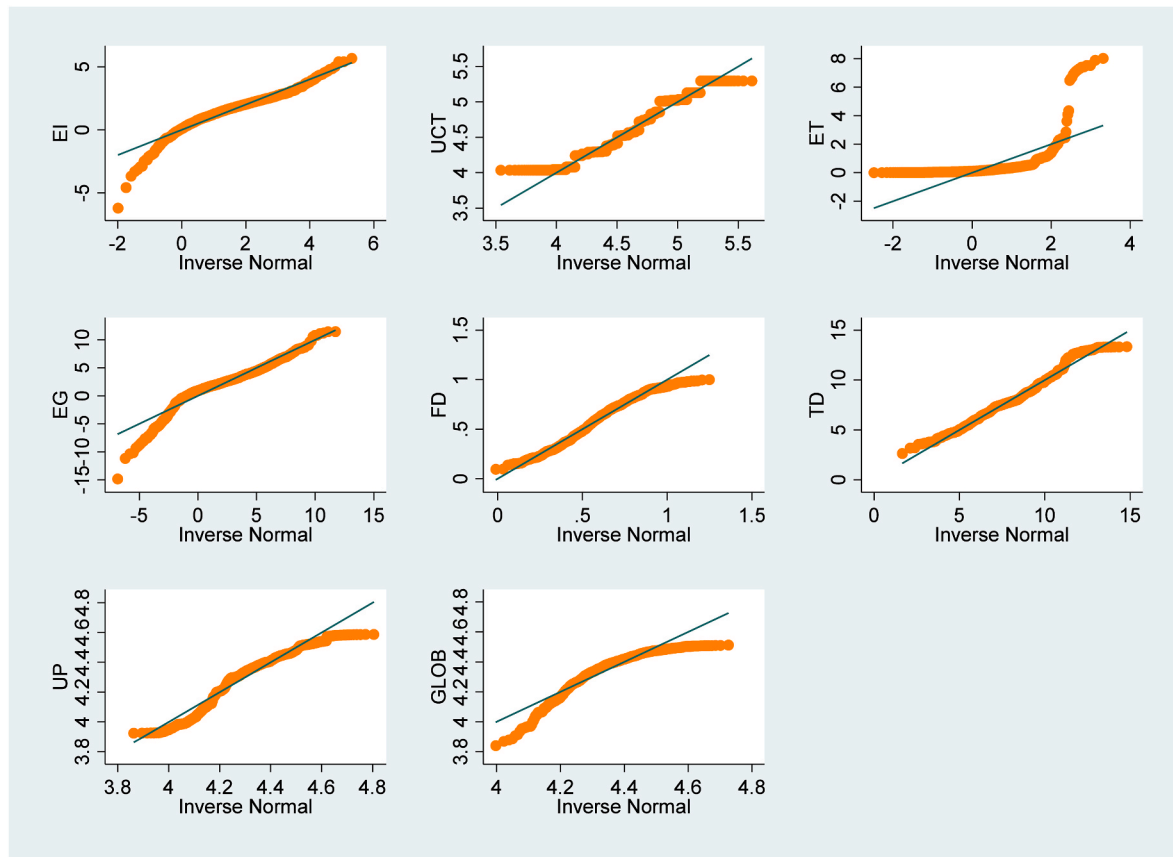
Table 1
Variables description.

Variables	Definitions	Sources
Energy inflation (EI)	Consumer price index: energy	OECD
US–China tension (UCT)	US–China tension index	Rogers et al. (2024)
Energy transition (ET)	The ratio between renewable energy consumption and fossil fuel energy consumption (% of total final energy consumption).	Authors' calculation
Economic growth (EG)	GDP growth (annual %)	WDI
Financial development (FD)	Financial development index	IMF
Technology development (TD)	Patent applications, total (residents and nonresidents)	WDI
Urban population (UP)	Urban population (% of total population)	WDI
Globalization (GLOB)	Globalization index	KOF

Table 2

Summary statistics.

	Mean	Median	Max	Mini	Std. Dev.	Skewness	Kurtosis	Jarque-Bera. [Prob.]
EI	1.662	1.790	5.665	-6.202	1.187	-1.074	7.571	0.000
UCT	4.578	4.531	5.296	4.035	0.337	0.289	2.157	0.000
ET	0.415	0.181	8.017	0.005	0.942	5.919	41.81	0.000
EG	2.452	2.607	11.46	-14.83	3.017	-0.974	6.592	0.000
FD	0.619	0.646	1.000	0.096	0.205	-0.299	2.204	0.000
TD	8.232	7.920	13.34	2.639	2.135	0.325	2.902	0.000
UP	4.334	4.365	4.587	3.924	0.153	-0.804	3.059	0.000
GLOB	4.362	4.393	4.512	3.840	0.118	-1.425	5.178	0.000

**Fig. 2.** Q-Q plots for normality.**Table 3**

Correlation matrix and VIF results.

	EI	UCT	ET	EG	FD	TD	UP	GLOB	Variable	VIF	1/VIF
EI	1								FD	2.75	0.364
UCT	0.072	1							GLOB	2.13	0.471
ET	-0.032	0.137	1						TD	2.02	0.496
EG	0.010	-0.222	-0.005	1					UP	1.37	0.730
FD	-0.138	0.145	-0.055	-0.186	1				ET	1.35	0.739
TD	-0.110	-0.018	-0.339	-0.063	0.508	1			UCT	1.25	0.800
UP	-0.081	0.135	0.216	-0.057	0.423	0.271	1		EG	1.09	0.917
GLOB	-0.121	0.369	-0.034	-0.228	0.556	-0.015	0.151	1	Mean VIF	1.71	

of I(0). Following the unit root analysis, the study proceeds to examine the long-run relationship among the variables through panel cointegration tests. The Kao and Pedroni panel cointegration tests are applied, and their results are presented in Table 6. Both tests strongly reject the null hypothesis of no panel cointegration, confirming the existence of a long-run equilibrium relationship among the variables.

The baseline results are obtained through four regression techniques,

namely, DK-FE, 2SLS, Lewbel 2SLS, and SGMM. The estimated results from all four regression techniques are reported in Table 7. The findings displayed in Table 7 confirm that UCT contributes positively to enhancing energy inflation in our analysis. The coefficients for UCT are significantly positive under 2SLS, Lewbel 2SLS, and SGMM regression techniques, showing that a 1 % rise in UCT would enhance EI by 0.735 %, 0.537 %, and 0.454 % under 2SLS, Lewbel 2SLS, and SGMM

Table 4
CD test and slope heterogeneity results.

	CD test		slope heterogeneity test	
	Pesaran's test	off-diagonal elements	$\hat{\Delta}$	$\hat{\Delta}_{adj}$
EI	38.02***	0.352	7.001***	8.430***
UCT	82.63***	0.672	18.07***	21.76***
ET	32.35***	0.473	4.480***	5.395***
EG	70.89***	0.573	−0.001	−0.001
FD	25.71***	0.406	20.56***	24.76***
TD	3.514***	0.507	19.32***	23.27***
UP	21.67***	0.549	24.80***	29.88***
GLOB	28.40***	0.443	17.13***	20.63***

Table 5
IPS test results.

	I(0)	I(1)
EI	0.021	−3.061***
UCT	−1.320	−6.548***
ET	0.343	−4.513***
EG	−4.748***	
FD	−2.726***	
TD	−0.943	−4.283***
UP	−6.181***	
GLOB	0.125	−3.079***

Table 6
Cointegration tests results.

		Statistic	p-value
Kao test	DF _t	−6.405***	0.000
	Modified DF _t	−6.653***	0.000
	Unadjusted DF _t	−11.34***	0.000
	Unadjusted modified DF	−21.86***	0.000
Pedroni test	ADF _t	−4.323***	0.000
	PP _t	−7.945***	0.000
	Modified PP _t	5.258***	0.000
	ADF _t	−8.128***	0.000

regression models, respectively. Although the UCT effect on EI is positive under DK-FE regression but remains statistically insignificant. Conversely, ET exerts a negative influence on EI in our analysis. The estimated coefficients for ET are significantly negative across the Lewbel 2SLS and SGMM estimation techniques, revealing that a 1 % increase in ET would mitigate EI by 0.138 % and 0.116 % in Lewbel 2SLS and SGMM models, respectively. Although the DK-FE and 2SLS models also report a negative association between ET and EI, this effect is statistically insignificant. EG, FD, and UP appear to have insignificant influence on energy inflation across all four models in our analysis. This suggests that variations in economic growth, financial development, and urban population have no significant impact on energy inflation in this model. The results further demonstrate that TD exhibits a declining effect on EI in this model. The coefficients for TD are significantly negative under the Lewbel 2SLS and SGMM estimation methods, indicating that a 1 % rise in TD reduces EI by 0.0892% and 0.0720%, respectively. While both the DK-FE and 2SLS models also show a negative relationship between TD and EI, the estimated effects in these models are statistically insignificant. Finally, GLOB exhibits a declining influence on EI in this model. The coefficients for GLOB are significantly negative under all four estimation methods, indicating that a 1 % rise in GLOB reduces EI by 4.366 % in DK-FE model, 2.457 % in 2SLS model, 2.057 % in Lewbel 2SLS model, and 1.571 % in SGMM model. Regarding diagnostic tests, the Hansen and AR(2) test results confirm instrument validity and the absence of serial correlation. The Kleibergen–Paap rk LM test indicates that the null hypothesis of under-identification is rejected, while the Kleibergen–Paap rk Wald F-statistic suggests that our instruments are relevant.

Table 7
Baseline results.

Variables	DK-FE	2SLS	Lewbel 2SLS	SGMM
	(1)	(2)	(3)	(4)
L.EI				0.448*** (0.0236)
UCT	0.734 (0.496)	0.735** (0.291)	0.537*** (0.128)	0.454*** (0.091)
ET	−0.0290 (0.169)	−0.102 (0.0684)	−0.138*** (0.0461)	−0.116** (0.0462)
EG	−0.0128 (0.0315)	0.0719 (0.160)	−0.0102 (0.0253)	0.0142 (0.0088)
FD	0.314 (1.323)	0.323 (0.512)	0.185 (0.304)	0.322 (0.365)
TD	0.0674 (0.0739)	−0.0648 (0.0396)	−0.0892*** (0.0248)	−0.0720** (0.0357)
UP	0.880 (1.873)	0.228 (0.515)	0.145 (0.286)	0.728 (0.473)
GLOB	−4.366*** (1.228)	−2.457*** (0.736)	−2.057*** (0.464)	−1.571*** (0.512)
Constant	12.83 (10.47)	10.20** (4.067)	9.505*** (2.278)	3.363 (2.744)
Observations	957	957	957	924
Number of countries	33	33	33	33
Hansen test [p-values]				0.245
AR(2) [p-values]				0.504
Kleibergen–Paap rk LM statistic [p-values]				0.025**
Kleibergen–Paap rk Wald [F-statistic]				32.54

Standard errors in parentheses.

***p < 0.01, **p < 0.05, *p < 0.1.

Considering the non-normal distribution and heterogeneity of variables, the study uses MMQR technique. Table 8 shows the outcome of MMQR technique. The location parameter shows the average effect of variables on EI. The results indicate that UCT has a positive and significant impact, meaning rising tensions increase EI on average. In contrast, ET has a negative and significant effect, suggesting that it helps reduce EI overall. Similarly, TD and GLOB also show negative and significant effects, indicating their role in lowering EI. Other variables such as EG, FD, and UP are mostly insignificant, meaning they do not strongly affect the average level of EI. The scale parameter reflects how these variables influence the variability or dispersion of EI. The results show that most variables have weak or insignificant effects on scale, except for some minor impacts. This suggests that while key variables affect the average level of EI, they do not significantly change its volatility across countries.

The quantile estimates reveal that UCT exerts a positive and statistically significant effect on EI across all quantiles except 0.10 quantile. The magnitude of the coefficient rises steadily from 0.419 at the 0.20 quantile to 0.753 at the 0.90 quantile, stating that the effect of UCT on EI strengthens as energy inflation exaggerates. This result strongly supported hypothesis 1. UCT shows a positive and significant impact on energy inflation across most quantiles, with the coefficient increasing from lower to higher quantiles. This confirms that geopolitical tensions exert stronger inflationary pressure when energy prices are already high, which aligns well with the theoretical expectations of risk premiums, supply disruptions, and precautionary demand. In contrast, ET demonstrates a significantly negative impact on EI across all quantiles, except 0.80 and 0.90 quantiles. The magnitude of the coefficients declines from −0.238 at the 0.10 quantile to −0.0928 at the 0.70 quantile, indicating that the mitigating effect of ET on EI is stronger at lower quantiles. Hypothesis 2 is also largely supported by the results. The ET shows a negative and significant effect across most quantiles, indicating its role in reducing energy inflation. However, the effect weakens at higher quantiles (0.80 and 0.90), which slightly contradicts the uniform expectation. This can be justified by the fact that during periods of

Table 8
MMQR results.

Variables	Quantiles										
	location	scale	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90
UCT	0.549*** (0.127)	0.145 (0.093)	0.319 (0.236)	0.419** (0.182)	0.477*** (0.154)	0.524*** (0.135)	0.571*** (0.121)	0.612*** (0.114)	0.651*** (0.113)	0.690*** (0.117)	0.753*** (0.133)
ET	-0.138** (0.0605)	0.0637 (0.0442)	-0.238** (0.112)	-0.194** (0.0862)	-0.169** (0.0731)	-0.148** (0.0643)	-0.128** (0.0576)	-0.110** (0.0542)	-0.0928* (0.0535)	-0.0754 (0.0554)	-0.0481 (0.0632)
EG	-0.0026 (0.0129)	0.0035 (0.0094)	-0.0083 (0.0238)	-0.0058 (0.0184)	-0.0044 (0.0156)	-0.0032 (0.0137)	-0.0021 (0.0123)	-0.0010 (0.0116)	-0.0001 (0.0114)	0.0008 (0.0118)	0.0023 (0.0135)
FD	0.195 (0.283)	-0.393* (0.206)	0.815 (0.523)	0.546 (0.403)	0.388 (0.342)	0.261 (0.300)	0.134 (0.269)	0.0229 (0.253)	-0.0808 (0.250)	-0.188 (0.259)	-0.357 (0.296)
TD	-0.0888*** (0.0235)	0.0336* (0.0171)	-0.142*** (0.0434)	-0.119*** (0.0334)	-0.105*** (0.0284)	-0.0944*** (0.0249)	-0.0836*** (0.0223)	-0.0741*** (0.0210)	-0.0652*** (0.0207)	-0.0561*** (0.0215)	-0.0417* (0.0245)
UP	-0.150 (0.268)	0.376* (0.196)	-0.743 (0.497)	-0.485 (0.382)	-0.334 (0.324)	-0.213 (0.285)	-0.0917 (0.256)	0.0146 (0.240)	0.114 (0.237)	0.216 (0.246)	0.378 (0.281)
GLOB	-2.033*** (0.438)	0.383 (0.320)	-2.637*** (0.809)	-2.374*** (0.625)	-2.221*** (0.530)	-2.097*** (0.466)	-1.974*** (0.417)	-1.865*** (0.393)	-1.764*** (0.388)	-1.660*** (0.402)	-1.495*** (0.458)
Constant	9.345*** (2.072)	-3.201** (1.512)	14.39*** (3.837)	12.20*** (2.949)	10.91*** (2.503)	9.878*** (2.201)	8.846*** (1.972)	7.940*** (1.855)	7.095*** (1.832)	6.221*** (1.898)	4.846*** (2.166)
Observations	957	957	957	957	957	957	957	957	957	957	957

Note: Robust standard errors in parentheses.

***p < 0.01, **p < 0.05, *p < 0.1.

extreme inflation, short-term transition costs, infrastructure constraints, and energy market rigidities may limit the stabilizing role of clean energy. Generally, the empirical findings are consistent with the theoretical framework. While minor variations across quantiles exist, they reflect distributional heterogeneity rather than inconsistency, thereby strengthening the validity of the proposed hypotheses.

In line with baseline regression results, the coefficients of EG, FD, and UP remain statistically insignificant across all quantiles, implying that variations in economic growth, financial development, and urban population do not have any significant effect on energy inflation in this model. TD displays a negative and statistically significant effect on EI across all quantiles. The magnitude of the coefficient declines from -0.142 to -0.0417 from 0.10 to 0.90 quantiles, describing that technology development helps reduce energy inflation across all quantiles, with a stronger effect observed at the lowest quantile. Finally, GLOB also exerts a negative and significant impact on EI across all quantiles. The magnitude of the coefficient decreases from -2.637 to -1.495 from 0.10 to 0.90 quantiles, indicating that globalization significantly helps in reducing energy inflation across all quantiles.

SIVQR regression technique is used to validate the results of the MMQR method (Table 8). SIVQR results are shown in Table 9. The SIVQR method provides nearly identical insights. Starting with UCT, Table 9 reveals that UCT significantly increases energy inflation from 0.60 to 0.90 quantiles, with its impact strengthening from 0.320 at the 0.60 quantile to 0.984 at the 0.90 quantile. Regarding ET, Table 9 shows a significantly negative association between ET and EI, reaffirming that energy transition tends to reduce energy prices from -0.155 at 0.40 quantile to -0.192 at 0.80 quantile. This finding implies that the benefits of energy transition in curbing energy inflation remain relevant in medium and high quantiles. Aligning with the results in Table 8, the effect of EG, FD, and UP remains negative but statistically insignificant across all quantiles in Table 9. The findings for TD remain consistent to Table 6. TD exerts a negative and significant effect on EI at middle and higher quantiles. In Table 9, the TD effect becomes statistically significant from the 0.50 quantile to the 0.90 quantile, reducing EI by -0.0961 at the 0.50 quantile and reaching -0.189 at the 0.90 quantile. Finally, the GLOB effect on EI remains significantly negative from 0.20 quantile to 0.70 quantile, confirming the positive contribution of globalization in reducing inflationary pressure.

The results of the MMQR and SIVQR are not identical and also show variation across quantiles in both magnitude and statistical significance. Nevertheless, this does not make our results less reliable; rather, it indicates implicit heterogeneity in the distribution of energy inflation. First and foremost, the signs of most variables, particularly the key variables, remain the same across both estimators. For instance, the impact of the energy transition is mainly negative across most quantiles in both MMQR and SIVQR; however, the stronger impacts are noticeable in the middle and upper quantiles. On the other hand, UCT show a positive and increasing impact across most quantiles in the MMQR, while they have an insignificant impact on energy inflation at lower quantiles but become strongly positive and significant at higher quantiles. This difference could be due to how MMQR and SIVQR treat the endogenous variable. The SIVQR explicitly corrects for endogeneity, whereas MMQR does not. As a result, SIVQR estimates are more reliable in the presence of endogenous regressors (such as energy transition), which explains the shifts in significance and coefficient size across quantiles. Although there are some numerical differences in the estimates of the MMQR and SIVQR, the economic interpretation remains largely consistent. The key factors, such as UCT and ET, show consistent signs, and the quantile-based approach provides more detailed information without contradicting baseline results. This confirms the robustness and reliability of our estimates. In general, the study's outcomes across all methods are mostly consistent in direction and significance across estimators, confirming the reliability of our findings. However, a small amount of variation in the estimates is expected, particularly when comparing estimates across different models, such as

Table 9
SIVQR results (robustness).

Variables	(0.10)	(0.20)	(0.30)	(0.40)	(0.50)	(0.60)	(0.70)	(0.80)	(0.90)
UCT	−1.051 (0.649)	−0.788 (0.466)	−1.164 (0.828)	0.0043 (0.476)	0.185 (0.147)	0.320* (0.176)	0.382* (0.222)	0.473* (0.245)	0.984*** (0.266)
ET	−0.203 (0.803)	−0.284 (0.289)	−0.147 (0.175)	−0.155** (0.0769)	−0.154** (0.0606)	−0.148* (0.078)	−0.172** (0.085)	−0.192** (0.096)	−0.033 (0.102)
EG	−1.034 (1.344)	−0.725 (0.477)	−1.008 (1.174)	−0.357 (0.459)	−0.291 (0.235)	−0.266 (0.530)	−0.305 (3.054)	−0.871 (11.11)	0.0112 (0.102)
FD	−2.506 (6.860)	−0.499 (3.329)	−1.155 (8.335)	−0.255 (3.414)	−0.171 (0.776)	−0.176 (1.850)	−0.344 (2.270)	−0.632 (2.557)	−0.557 (0.887)
TD	0.0255 (0.979)	−0.112 (0.217)	−0.213 (0.272)	−0.101 (0.123)	−0.0961* (0.0521)	−0.0932* (0.0543)	−0.0913* (0.0474)	−0.176*** (0.0509)	−0.189*** (0.0649)
UP	1.603 (1.374)	−0.0721 (1.055)	0.700 (1.772)	0.0991 (1.267)	0.0882 (1.232)	0.112 (0.513)	0.142 (0.331)	0.259 (0.415)	0.277 (0.497)
GLOB	0.366 (2.020)	−3.427** (1.719)	−4.942** (1.950)	−3.172* (1.640)	−3.082** (1.257)	−3.027*** (1.142)	−3.126* (1.892)	−4.072 (3.249)	−1.707 (1.052)
Constant	7.194 (20.20)	22.14 (27.63)	44.49 (36.88)	16.30 (13.43)	15.59*** (5.984)	15.41*** (5.840)	15.80 (19.92)	40.97 (31.61)	5.093 (8.651)
Observations	957	957	957	957	957	957	957	957	957

Note: Robust standard errors in parentheses.

***p < 0.01, **p < 0.05, *p < 0.1.

mean vs quantile estimation, static vs dynamic frameworks, and the treatment of endogeneity.

The large negative coefficients of GLOB make sense from an economic point of view. Globalization significantly enhances trade liberalization, energy supply efficiency, and market competition, all of which play a critical role in lowering energy costs. This is why its increasing size shows how important it is for stabilizing global energy markets. Differences amongst estimators is also expected. GLOB has consistently detrimental and substantial impacts in DK-FE and IV models. In MMQR, the impact stays negative for all quantiles, but it is smaller, which means that it has less of an effect when inflation is greater. In SIVQR, some variation occurs because of endogeneity correction. Overall, the sign of the estimates stays the same, which shows the reliability and robustness of our results despite differences in methods used.

4.1. Discussion

Firstly, the outcomes of the analysis highlight the significance of US-China tensions in enhancing energy inflation. The findings provide strong support for Hypothesis 1. This occurs because escalating tensions between the US and China lead to increased risk premiums on oil and gas, which in turn hinders trade and causes energy costs to be transmitted more fully to consumer prices. The empirical evidence in this regard underscores that increased geopolitical risks, due to rising tensions between the US and China, lead to enhanced restrictions on energy markets and elevate inflation via both supply and demand mechanisms. The empirical results are broadly in line with past research on the geopolitical risks and energy markets. The study reveals that the rising US-China tensions have a significant effect on energy inflation in most quantiles. The result corroborates the previous research by [Cai et al. \(2022\)](#), [Li et al. \(2025\)](#), and [Mignon and Saadaoui \(2024\)](#), which showed that geopolitical tensions bring uncertainty, cause the breakage of supply chains, and result in volatility in energy markets. The greater effect on higher quantiles also indicates that the geopolitical tensions worsen in times of already high energy inflation. The prior literature does not fully support our study results. According to [Li et al. \(2025\)](#), as tensions between the US and China increase, the prices of fossil fuels become more volatile, thereby enhancing the operational costs for transportation, energy, and industry. The study further confirms that US-China tensions adversely influence the oil and gas industry. Since energy shocks have the ability to increase global energy prices, instability in the energy market has significant implications for energy inflation. [Cai \(2025\)](#) observed that rising US-China tensions proved detrimental to the global supply chain and economy, ultimately increasing the general price level. [Cai et al. \(2022\)](#) demonstrated that

US-China tension is adversely affects the oil market. As a result of these tensions, both nations experienced a decline in their energy demand and an overall economic contraction. [Mignon and Saadaoui \(2024\)](#) also support our findings, stating that US-China tensions enhance geopolitical risks, which ultimately lead to increased energy prices. The rising impact of US-China tensions on energy inflation across quantiles reflects different market conditions. At lower quantiles, energy prices are relatively stable, supply is sufficient, and markets can absorb geopolitical shocks with limited price changes. As energy inflation moves to higher quantiles, supply is constrained, and uncertainty increases, significantly increasing the sensitivity of energy prices to external shocks. In such conditions, US-China tensions enhance the trade and shipping costs, upset energy supply chains, and surge risk premiums in energy markets. Similar empirical evidence has been provided by [Zaghdoudi \(2026\)](#). When inflation is already high, the above-stated effects are more easily passed on to prices. In addition, uncertainty discourages energy investment and weakens currencies in importing countries, further pushing up energy costs.

Second, energy transitions play a critical role in reducing energy inflation. This empirical finding confirms Hypothesis 2. The biggest trigger of energy inflation is the susceptibility of the fossil fuel market to rising geopolitical risk, which undermines its supply and thereby raises energy prices. However, renewable energy, on the other hand, is less likely to be affected by such risks due to its decentralized nature and indigenous production. Unlike traditional fossil fuels, which are concentrated in certain parts of the world, such as the Middle East, Central Asia, Russia, China, and America, renewable energy can be produced anywhere in the world where there is an ample supply of sun, wind, or water. Therefore, to counter the negative impacts of rising energy inflation on the global economy, nations are transitioning towards clean and low-carbon energy sources. The findings are partially consistent with other research, such as that by [Jin and Kim \(2023\)](#) and [Cevik and Ninomiya \(2023\)](#), which state that renewable energy bolsters energy security and lessens reliance on volatile fossil fuel markets. This effect, however, diminishes at higher quantiles, suggesting that the stabilizing role of renewable energy can be weakened during periods of extreme inflation due to other factors such as short-run transition costs, infrastructure constraints, and market rigidities. The study thus corroborates previous findings and also deepens the understanding of these relationships by revealing that they are different depending on the level of energy inflation. The study by [Jin and Kim \(2023\)](#) indicated that, in addition to decarbonizing the energy industry, the implementation of clean energy sources can enhance energy security from an economic perspective. Thus, the transition towards low-carbon energy alternatives may serve as a hedge that mitigates adverse effects on the economy. Our

empirical findings align with those of [Jin and Kim \(2023\)](#). Transitioning to renewable energy promotes economic development and offers the added advantage of energy security. The efficacy of low-carbon and clean energy on economic development is also confirmed by the study of [Fankhauser and Jotzo \(2018\)](#), which ultimately brings stability in the economy by controlling inflation. Thus, the negative connection between energy transition and energy inflation is justified.

Our results show that the energy transition has not fully stabilized energy prices; instead, the results show that early-stage energy transition efforts offer some sort of stabilization, particularly during times of low to moderate inflation. Since the transition is still in its early and developing phase, increased renewable capacity, and reduced dependence on fossil fuel imports can slow but surely reduce the exposure to geopolitical shocks. Thus, energy transition contributes to price stability over time ([Cevik and Ninomiya, 2023](#)), although its impact remains limited during periods of extreme inflation or major external shocks. The results also show that the energy transition helps reduce energy inflation more pronouncedly when prices are low to moderate. At lower to middle quantiles, enhancing green energy, improving efficiency, and reducing fossil fuel dependence play a critical role in stabilizing energy supply and lowering exposure to global price shocks. As energy inflation moves to higher levels, energy markets become more volatile, with disrupted supply conditions and short-term constraints. At the highest quantiles, energy inflation is usually driven by sudden shocks like geopolitical tensions, supply disruptions, or extreme weather. In these situations, immediate supply shortages and higher risk premiums dominate energy prices, which weakens and often eliminates the visible impact of the energy transition.

5. Conclusion and implications

Geopolitical risk and energy prices are closely linked. With rising tensions between the US and China, the world's two largest economies, the issue of energy inflation has once again gained attention. Despite growing concerns about environmental sustainability, fossil fuels remain the primary source of energy consumption worldwide. The rising tensions between the US and China have escalated geopolitical tensions, hindered supply, and impacted the prices of fossil fuels. Since fossil fuels are vulnerable to supply shocks in global energy markets, policymakers worldwide are seeking alternative options that can be produced domestically and are less susceptible to rising geopolitical risks. Therefore, it is crucial to understand the role of rising US-China tensions in shaping the dynamics of energy inflation and whether the transition to clean and green energy sources can bring stability to energy prices. Against this backdrop, the primary objective of the analysis is to investigate the impact of US-China tensions and energy transitions on energy inflation, employing the MMQR estimator. Previous studies primarily examined the effects of geopolitical risks on energy markets. The prior literature also ignores the impact of energy transition on energy inflation.

Therefore, our study is important in providing evidence of the different effects of geopolitical tensions and energy transition on energy inflation at different levels of inflation. While energy transition brings down the energy inflation, such a system is less effective when energy inflation is extremely high, proving that there are still structural and infrastructural constraints on renewable energy systems. One of the key novelties of this study is the analysis of MMQR and SIVQR methods, which show the effects are heterogeneous and asymmetric, and cannot be captured by traditional mean-based methods. These approaches give only an average effect, and this approach provides the mechanism for the effects of US-China tensions and energy transition to vary between low and high inflation scenarios. As such, the study also makes a methodological contribution since it provides a more detailed understanding of the dynamics of energy inflation and more realistic policy findings for controlling inflation in the face of geopolitical uncertainty. The outcomes confirm that US-China tensions and energy inflation are

positively correlated at all quantiles, implying that rising US-China tensions enhance energy inflation at every level. On the other hand, the findings show that energy transition reduces energy inflation at low and medium levels. Thus, we can state that the energy transition provides stability effects to higher energy inflation, but not when the energy inflation is too high.

Considering the significance of the outcomes, we have proposed the following policy recommendations. First, the results indicate that the impact of US-China tensions on energy inflation intensifies as it moves from the lower to the higher quantiles, from 0.419 at the 0.20 quantile to 0.753 at the 0.90 quantile. This suggests that US-China tensions generate a lot more inflationary pressure at times of high energy inflation. As a result, the policy makers must decrease reliance on imported fossil fuels, diversify energy sources with different countries, improve energy cooperation within the region, and establish strategic energy reserves for less impact of geopolitical issues on the domestic energy market. The results also reveal that energy transition has a clear impact on reducing energy inflation for the lower quantiles (0.10 to 0.70), and the more stabilizing effect is seen at lower inflation levels. This indicates that the policies for renewable energy are more effective when energy inflation has not reached an extreme level. Governments thus need to boost investments in renewables infrastructure, offer incentives and subsidies to encourage the use of renewables, and encourage long-term energy transition plans to lessen dependence on oil and gas markets and enhance energy price stability. The recommendations for policy are based directly on the empirical results of the study. It is revealed that the US-China tension exerts a significant effect on almost all quantiles of energy inflation, with greater effects at higher inflation levels. Likewise, the results show that the energy transition also proves to be very effective in lowering energy inflation from low to medium quantiles.

Due to the differences in the magnitude of the effects of US-China tensions and energy transition on energy inflation across its various levels, we observed an asymmetry in these effects. Thus, it is suggested that policymakers focus on these varying effects when dealing with low, medium, and high levels of energy prices. The rise in US-China tensions has significantly increased energy prices, suggesting that policymakers should focus on the production of indigenous and alternative energy sources. Although the geopolitical risks are uncontrollable and can't be fully avoided, their impacts can be minimized by enhancing cooperation with energy-rich countries. It is suggested that energy-exporting and importing nations collaborate with global powers to sign a deal that enhances the safety and security of trade routes and ensures the smooth supply of energy sources. The forward purchase of oil and gas could also be an option to monitor rising energy prices in the wake of global tensions. We observed that the energy transition lowers energy prices. Thus, we suggest that policymakers should enhance the production of low-carbon energy sources that are less vulnerable to any supply shock due to their decentralized nature and local production. Government needs to allocate more funds to invest in research and development activities for the development of low-carbon energy technologies. Governments should design policies that make the environment more conducive to investment in high-risk clean energy ventures by offering subsidies, tax rebates, and low-interest loans that would attract a large number of private investors.

Since the impacts of US-China tension are increasing on energy inflation as we move from the lower to the highest quantiles, we suggest that the government should also adopt different policies depending on the level of energy inflation. In times of low inflation, governments should focus on diversifying energy import sources, expanding strategic reserves, improving energy-market data, and encouraging long-term supply contracts. On the other hand, as inflation rises, policymakers should activate stronger protections by offering temporary support for vulnerable households and targeted relief for critical industries. In addition, the government should also keep an eye on the rising energy prices and speculations that push energy prices upwards. To reduce high-quantile vulnerability, it is suggested to invest in domestic energy

security by enhancing the generation of clean and green energy sources, enhancing the capacity of storage facilities, building grids, and upgrading the transmission channels. Further, the governments should boost regional energy cooperation and should look for alternative trading routes and supplies in case of supply and trade disruptions. This reduces risk premiums during US-China geopolitical shocks. In contrast, energy transition helps reduce the energy inflation from the lowest to the highest levels. When the energy inflation is low, governments should focus on saving energy and should also support long-term renewable energy contracts that keep prices stable and predictable. As inflation rises, short-term problems like limited grids and storage reduce these benefits, so policymakers should invest in strengthening energy systems to handle renewables better.

Appendix

Table A1
List of countries

Australia	Netherlands	Iceland
Spain	Lithuania	Norway
Poland	USA	Germany
Canada	Finland	Austria
Sweden	Switzerland	Czechia
Portugal	Slovak Republic	UK
New Zealand	Belgium	Colombia
Denmark	Slovenia	Mexico
France	Japan	Israel
Hungary	Korea, Rep.	Türkiye
Italy	Luxembourg	Greece

Data availability

Data will be made available on request.

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CRedit authorship contribution statement

Ebrahim Merza: Conceptualization, Data curation, Formal analysis, Investigation. **Ilhan Ozturk:** Methodology, Resources, Software, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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